

Problem 9

Evaluate the integral:

$$I = \int \frac{x^2}{1+x^2} dx.$$

Solution

To solve the given integral, we decompose the fraction and simplify it step by step.

Step 1: Simplify the integrand

Rewrite the numerator x^2 as:

$$x^2 = (1+x^2) - 1.$$

Substituting this into the integrand:

$$\frac{x^2}{1+x^2} = \frac{(1+x^2) - 1}{1+x^2} = 1 - \frac{1}{1+x^2}.$$

Thus, the integral becomes:

$$I = \int \left(1 - \frac{1}{1+x^2} \right) dx.$$

Step 2: Split the integral

Split the integral into two simpler integrals:

$$I = \int 1 dx - \int \frac{1}{1+x^2} dx.$$

Step 3: Evaluate each integral

1. The first integral is straightforward:

$$\int 1 dx = x.$$

2. The second integral, $\int \frac{1}{1+x^2} dx$, is a standard integral:

$$\int \frac{1}{1+x^2} dx = \arctan(x).$$

Step 4: Combine the results

Combine the results of the two integrals:

$$I = x - \arctan(x) + C,$$

where C is the constant of integration.

Final Answer

The solution to the integral is:

$$I = x - \arctan(x) + C.$$